

Stale Coherence: Quantifying Verbal–Action Dissociation in Long-Horizon Structured LLM Output

(with Project-Utopia, a deterministic measurement substrate)

Anonymous Authors¹

¹Affiliation withheld for double-blind review

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Abstract

A production-deployed LLM agent rarely fails by forgetting an anchor outright. It fails by a more subtle failure mode: it continues to *mention* the anchor in its summaries while its directives stop *acting* on it. We call this failure mode **stale coherence** and argue that it is an independent axis of LLM behavior distinct from raw recall as measured by LongMemEval [24], RULER [9], Lost-in-the-Middle [13], or MemoryArena [8].

We define stale coherence operationally as the dual probe ($\text{verbal_recall}(t)$, $\text{action_grounded_recall}(t)$) and the dissociation metric $\Delta(t) = \text{verbal_recall}(t) - \text{action_grounded_recall}(t)$. A single anchor is injected at $t = 0$ carrying both verbal tokens (“warehouse at (12, 8)”) and an implicit action goal (`{deliver, warehouse}`). At every sample point $t > 0$, the *verbal* probe checks whether the LLM-emitted summary still mentions the anchor; the *action-grounded* probe checks whether the LLM’s structured directives still allocate non-trivial weight to the anchor’s implicit goal. Stale coherence is the regime where these two probes diverge.

To make stale coherence measurable, we build Project-Utopia, a deterministic 8-sim-hour colony-simulation harness in which a single LLM emits structured directives across four parallel channels (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) at three nested cadences. The multi-cadence structured-output substrate is what makes the dissociation visible: single-cadence benchmarks cannot dissociate verbal mention from sustained behavioral commitment. Schema validation and idempotent guardrails ensure that what we measure is reasoning drift, not parse error or hardware non-determinism (three-tier reproducibility with bit-identical fallback hashes `be19781c...` verified at seed `0xC0FFEE`).

We evaluate a single pre-registered hypothesis: **H**: for every LLM family m in our model menu, there exists $t^*(m) \leq 4$ sim-hours such that $\Delta(t^*) > 0.3$. We additionally test whether stale coherence is statistically independent of LongMemEval-style verbal recall: if the dissociation gap correlates strongly with raw recall, the phenomenon reduces to known long-context bias; if it does not, stale coherence is a new axis production agent designers must explicitly test for.

At submission time, the pipeline is verified end-to-end on fallback-only NDJSON. Real-LLM evaluation across Claude-Sonnet-4.6, GPT-5-mini, and Llama-3.1-8B is reserved for the camera-ready once API access is provisioned. Project-Utopia is released as a Python package (`pip install project-utopia`, 615 tests passing, ~ 23 k LOC, 128 modules) at [ANONYMIZED-REPO-URL] (de-anonymized for camera-ready).

1 Introduction

A production-deployed LLM agent rarely fails by forgetting an anchor outright. It fails by a more subtle failure mode: the model continues to *verbally* mention an anchor — “warehouse at (12, 8)”, “defend territory X ”, “the patient is on warfarin” — in its running summaries and

reflections, while its *action-emitting* directives quietly stop reflecting it. The summary still says “I should defend territory X ”; the next dispatched unit goes elsewhere. The reflection still references the warehouse; the next pick task allocates zero weight to deliveries there. We call this failure mode **stale coherence**.

Stale coherence is the failure mode that breaks long-running agent deployments, and yet it is invisible to the dominant evaluation regimes. Existing memory benchmarks measure only one axis of the phenomenon. We argue this is a single, operationally definable phenomenon whose joint behaviour distinguishes it from every individual axis those benchmarks measure.

Existing benchmarks measure only one axis.

- **Verbal-recall benchmarks.** LongMemEval [24], RULER [9], Lost-in-the-Middle [13], and InfiniteBench [27] measure whether the model can *retrieve* a fact from long context, not whether it sustains *action* on it. Their probe is verbal: “what was X ?”
- **Action-grounded benchmarks (session-discrete).** MemoryArena [8] and MemoryAgentBench [10] do score action grounding, but in session-discrete agent loops: each task is bounded, the agent has explicit reset boundaries, and the action set is small. They cannot expose a drift that emerges only across continuous tick-by-tick context accumulation.
- **Production agent observations.** FAIR Cicero [15] and Anthropic’s multi-agent research system [2] anecdotally describe stale coherence (“the agent kept saying it would defend territory X but stopped moving units there”; inter-channel coherence loss in long-running sessions), but provide no controlled measurement.

No existing benchmark dissociates verbal mention from sustained behavioural commitment on tick-continuous, naturally-accumulated context. The phenomenon falls in the gap between three literatures and consequently has neither an operational definition nor a measurement instrument.

Contributions. We make three contributions, organized phenomenon-first:

C1 — Phenomenon. An operational definition of stale coherence as the dual probe (verbal_recall(t), action_grounded_recall(t)) with dissociation metric $\Delta(t) = \text{verbal_recall}(t) - \text{action_grounded_recall}(t)$. Stale coherence is distinct from forgetting (both axes can remain non-zero), distinct from positional bias (the anchor is fixed at $t = 0$ and probed at the same surface continuously), and distinct from session-discrete action grounding (the agent never receives a query about the anchor; behavioural commitment is measured implicitly across emission cadence).

C2 — Method. Anchor injection at $t = 0$ with a paired (verbal token, implicit action goal) — e.g. (“warehouse at (12,8)”, {deliver, warehouse}); a multi-cadence structured-output substrate (four parallel decision channels emitting at three nested cadences over a fixed four-tool surface) as the elicitation harness; and 8-sim-hour episodes matching the natural TTL spread of agent directives (8–24 h), which the route-B 240-sim-second framing did not exercise. The substrate is load-bearing methodologically because single-cadence benchmarks cannot dissociate verbal mention from sustained behavioural commitment.

C3 — Empirical (camera-ready). Cross-family decay profiles of $\Delta(t)$ over 8 sim-hours, and a statistical-independence test of Δ against LongMemEval-style verbal-only recall, across at least three LLM families (Claude-Sonnet-4.6, GPT-5-mini, Llama-3.1-8B).

Single pre-registered hypothesis. We register one testable hypothesis driving the entire empirical contribution: **H**: for every LLM family m in our model menu, there exists $t^*(m) \leq 4$ sim-hours such that $\Delta(t^*) > 0.3$. We additionally test whether Δ is statistically independent of LongMemEval-style verbal recall: Spearman correlation between the dissociation gap and

verbal-only recall < 0.5 . A correlation above this threshold would reduce stale coherence to known long-context bias; a correlation below it establishes stale coherence as a separate axis production agent designers must explicitly test for.

Why stale coherence dissociates from existing memory benchmarks. LongMemEval and RULER measure single-axis verbal recall; MemoryArena and MemoryAgentBench measure action grounding but in session-discrete tasks with explicit query-response boundaries. Stale coherence is the *joint* behaviour of both axes under tick-continuous long-horizon context, and it can appear even when each axis alone looks healthy: verbal_recall(t) near 1 (the model still talks about the anchor) and action_grounded_recall(t) near 0 (no directive weights it). The theoretical underpinning for the decay shape itself is the diminishing-returns geometry described by Sinha et al. [21]: recall scores decay sub-linearly with effective context length; stale coherence is the operational counterpart for the *action* axis and is observable only when the two axes are measured jointly on naturally-accumulated context.

Roadmap. Section 2 surveys memory benchmarks along the two axes above and the production-agent failure literature; Section 3 describes the multi-cadence structured-output substrate that makes the dual probe measurable; Section 4 presents the dual probe methodology and supporting scoring stack; Section 5 reports the cross-family decay experiments; Section 6 discusses implications for production agent design; Section 7 states limitations; Section 8 describes the reproducibility package.

2 Related Work

Stale coherence sits in a gap between two literatures: long-context and memory benchmarks (§2.1), which measure verbal or action recall but not their joint behaviour on tick-continuous context, and production-agent failure observations (§2.2), which anecdotally report the phenomenon but do not measure it. We organize this section accordingly.

2.1 Long-Context and Memory Benchmarks

Verbal-recall axis. The dominant long-context memory regime probes verbal retrieval: **LongMemEval** [24] curates multi-session conversational memory queries; **RULER** [9] synthesizes needle-in-a-haystack and aggregation tasks across context lengths; **Lost-in-the-Middle** [13] characterizes the U-shaped position bias of retrieval over long inputs; **InfiniteBench** [27] pushes the same probe beyond 10^5 -token contexts. HELM’s recent long-context battery [22] aggregates these into an omnibus verbal-recall score. In each case the figure of merit is “did the model retrieve the fact?”; none ask “did it *act* on it?”. As a result a model that hallucinates an anchor in its summary while ignoring it in its directives is indistinguishable from one that genuinely sustains commitment.

Action-grounding axis (session-discrete). A second strand evaluates whether the agent *acts* on remembered facts: **MemoryArena** [8] probes action grounding by issuing explicit downstream queries that require recalled facts; **MemoryAgentBench** [10] adds selective forgetting and structured tasks. These are the only benchmarks that touch the action axis at all, but both are *session-discrete*: each task has explicit boundaries, the agent receives an explicit query that re-cues the fact, and the action set is small. Stale coherence, in contrast, requires *tick-continuous* naturally-accumulated context: the agent emits structured directives continuously, never receives an explicit query about the anchor, and decay is measured by comparison across time rather than by query-response matching. The two regimes probe different failure modes —

session-discrete action grounding asks whether the agent retrieves on demand; stale coherence asks whether it sustains commitment when no demand is issued.

Theoretical underpinning. Sinha et al. [21] characterizes long-context recall as following a diminishing-returns curve: per-token information yield decays sub-linearly with effective context length. We adopt this geometry as the expected functional form for both axes of the dual probe and report decay profiles for $\Delta(t)$ rather than scalar end-of-episode scores. The contribution of this paper is the operational counterpart of that theory for the *action* axis, and the demonstration that the gap between the two axes is itself a measurable independent phenomenon.

Position. Stale coherence is the dissociation visible only when both axes are measured jointly on tick-continuous context. No existing benchmark probes both axes simultaneously, and no existing benchmark runs long enough on naturally-accumulated agent context for the dissociation to emerge. The methodological apparatus of §4 — dual probe, anchor injection at $t = 0$, 8-sim-hour episodes — is built specifically to close this gap.

2.2 Production Agent Failure Observations

The phenomenon we name has been reported anecdotally in production agent deployments. **FAIR Cicero** [15], the Diplomacy negotiation agent, informally documents cases in which the system continued to declare strategic intent (“defend territory X ”) in its dialogue generations while its unit-movement directives shifted away from that territory — a textbook instance of verbal-action dissociation in a production system. Anthropic’s multi-agent research system [2] reports inter-channel coherence loss in long-running sessions of its production multi-agent deployment, including cases where summary-channel outputs continued to assert commitments that the acting-channel outputs no longer reflected.

These reports are anecdotal, qualitative, and unreproducible: they describe the failure mode but do not measure it, isolate it from raw recall, or quantify its prevalence across model families. The contribution of this paper, relative to these observations, is to take the same phenomenon and deliver controlled, quantified, reproducible measurement: a single anchor, two paired probes, an 8-sim-hour deterministic substrate, a single pre-registered hypothesis, and three model families.

3 Project-Utopia Architecture

Project-Utopia is the elicitation substrate that makes stale coherence measurable. To dissociate verbal mention from sustained behavioral commitment we need (i) continuous structured-output emission across multiple temporal scales, (ii) a deterministic environment whose state-references can be machine-checked against the LLM’s directives, and (iii) an anchor injection protocol that pairs verbal tokens with implicit action goals. We satisfy these three requirements with a $4 \text{ channel} \times 4 \text{ tool}$ substrate over a seeded, bit-identically reproducible simulator. The substrate is the *elicitation method*, not the contribution of the paper; the contribution is the dual-probe characterization of stale coherence that it makes possible (§4).

3.1 Multi-Cadence Substrate

The substrate emits structured output across three nested temporal scales through four parallel LLM decision channels, each bound to one deterministic tool. The cadence spread is what dissociates verbal mention from behavioral commitment: a single-cadence harness cannot tell whether an LLM continues to *summarize* an anchor after it has stopped *acting* on it.

[Figure 1 placeholder] 4 channel $\times$ 4 tool elicitation substrate; three nested cadences (5–15s, 90s, sim-time gated); anchor injection at $t = 0$ via `world_summary`; dual-probe taps on `primaryGoal/constraints` (verbal) and `targetPriorities/intentWeights` (action).

Figure 1: Project-Utopia as the elicitation substrate for stale coherence. Multi-cadence structured output is what makes verbal-action dissociation visible; the four tools are kept fixed so the substrate is deterministic.

Environment-director (cadence: 5–15s; tool: event scheduler). Schema: {weather: enum `clear|rain|storm`, `durationSec` $\in [8, 180]$, `factionTension` $\in [0, 1]$, `eventSpawns`: ≤ 3 with intensity ≥ 0.4 }. Coarse-cadence state perturbations to which the policy channel must adapt, matching the ReAct [25] observe $\rightarrow$ act loop.

NPC-policy (cadence: 5–15s; tools: A* path query, Boids steering, FSM intent applier). Schema: {`groupId`, `intentWeights` $\in [0, 3]^{|T_g|}$, `riskTolerance` $\in [0, 1]$, `targetPriorities` $\in [0, 3]^{|T_g|}$, `ttlSec` $\in [8, 24\text{h}]$ }. Intent weights flow through `POLICY_INTENT_TO_STATE` into per-NPC FSM preferences; `targetPriorities` drive A* target selection; group Boids steering consumes `intentWeights`. The `targetPriorities` and `intentWeights` channels carry the *action-token sets* that the action-grounded recall probe (§??) machine-checks against injected anchors.

Strategic-plan (cadence: 90s heartbeat + crisis triggers; tool: plan arbitration). Schema: {`primaryGoal`: str, `constraints`: str[], `resourceBudget`: {food, wood, stone}}. The longest-cadence channel carries the *verbal summaries* in `primaryGoal/constraints` that the verbal-recall probe scans for anchor token mentions.

Colony-agent (cadence: sim-time-gated + plan-stall triggers; tool: build planner over PlanExecutor). Schema: [(`action`: enum, `args`: obj, `depends_on`: int[])]. Wraps a Perceive $\rightarrow$ Plan $\rightarrow$ Ground $\rightarrow$ Execute $\rightarrow$ Evaluate $\rightarrow$ Reflect loop [19, 25] that grounds multi-step build plans tile-by-tile.

3.2 Anchor Injection Protocol

At $t = 0$ we inject **5 anchor entries** into the LLM’s working memory via the system prompt’s `world_summary` field. Each anchor carries two paired fields:

Listing 1: Anchor entry shape

```
{
  "verbal_tokens": ["warehouse at (12,8)"],
  "implicit_goals": [
    {"action": "deliver", "target": "warehouse"}
  ]
}
```

The two fields are deliberately *not* co-presented as an instruction (“you must deliver to (12,8)”). The anchor appears as observed world state. The LLM must autonomously integrate it into its directive stream: verbal tokens may resurface in `primaryGoal/strategy.notes`, and implicit goals may be served by allocating non-trivial weight to {`action:deliver`, `target:warehouse`} in the `targetPriorities/intentWeights` channels. The two paths are independent: an LLM may surface verbal tokens without serving implicit goals (stale coherence), serve implicit goals without surfacing verbal tokens (silent commitment), or do both (coherent) or neither (forgotten).

The 5-anchor count is empirically motivated. In pilot runs (Appendix E) the verbal-recall probe saturates near 1.0 with < 5 anchors (capacity not stressed) and the instruction-following

probe collapses for the smaller open-weight models with > 5 anchors (capacity exceeded); $n = 5$ is the elbow at which both probes have measurable variance across our model menu.

3.3 Schema Validation and Guardrails

The substrate must distinguish reasoning drift from parse error. Two filters sit between every LLM emission and the tool layer. **ResponseSchema** performs JSON-Schema validation (unknown enums dropped, NaN rejected, types coerced). **Guardrails** clamps numeric ranges (weights $\in [0, 3]$, risk $\in [0, 1]$, TTL $\in [8\text{h}, 24\text{h}]$) and is an idempotent fixed point: `guard(guard(x)) == guard(x)` for every directive shape (generalizing the typed-function-signature pattern of ToolLLM [17] and Gorilla [16] to four parallel channels). Idempotency matters here because clamped output from one channel re-enters another channel’s prompt on the next tick; a non-idempotent guard would compound rather than stabilize. In the stale-coherence context the schema layer is *plumbing, not the headline*: its role is to provide machine-checkable ground truth for whether a directive “references an anchor” or not, so that the dual probes (§??) measure reasoning behavior rather than JSON malformedness.

3.4 Determinism Seam

Without bit-identical reproducibility we cannot distinguish stale coherence from hardware variance. RNG flows through `project_utopia.app.rng`; A* paths are cached by (faction, gridVersion, costVersion); Boids accumulators run in fixed order; PathWorkerPool is disabled in `deterministic=True` mode. `project_utopia.tools.audit.determinism_check` runs the harness twice on the same seed and compares SHA-256 hashes of slim state snapshots; three tiers pass with bit-identical hashes: 30 ticks `be19781c...`, 1 800 ticks `f7a099a4...`, 7 200 ticks `43cb6aec...` (Appendix A).

The **AgentAdapter** is the interface any policy (LLM, fallback, scripted oracle) implements to drive the four channels:

Listing 2: AgentAdapter contract

```
async def request(self, channel: str, payload: Any,
                  options: dict | None = None) -> DecisionResponse:
    # channel in {environment-director, npc-policy,
    #             strategic-plan, colony-agent}
    # MUST NOT raise; on failure return a fallback DecisionResponse
    ...
```

Concrete adapters: **NoopAgentAdapter** (sandwich-norm lower bound), **LLMClient** (OpenAI-compatible HTTP via litellm), **FlatBaselineAdapter** (single-channel control that emits per-worker FSM-state preferences directly), and **ScriptedOraclePolicy** (hand-tuned upper bound). Tier-2 deterministic-inference adapters and cassette replay are documented in §8. The per-channel table mapping each directive schema to its tool implementation is deferred to Appendix E; it is supporting machinery for the dual probe, not the central object of study.

4 Metric Stack

Project-Utopia reports scores at four layers, each grounded in prior art (Figure 2).

4.1 Layer 1: Per-Tick Dimension Plugins

Five plugins (`project_utopia/benchmark/dimensions/`) emit a per-cell score row of 19 keys:

[Figure 2 placeholder] 4-layer metric stack: Per-tick raw signal → Sandwich norm + Geometric mean → Bayesian per-dimension → HELM MWR cross-model leaderboard.

Figure 2: Four-layer metric stack. Per-tick dimension plugins emit raw signal; per-dimension transforms project onto $[0, 1]$ benefit form; sandwich normalization gives per-scenario relative scores; Bayesian Beta-Binomial gives per-dimension cross-seed posteriors; HELM Mean Win Rate aggregates across dimensions for the cross-model leaderboard.

ResourceAllocationEfficiency ($N=5$): `rae_composite`, `rae_sufficiency`, `rae_distribution_gini`, `rae_idle_capacity`, `rae_path_overhead`. `rae_composite` is the Crafter geometric mean [7] over per-resource sufficiency:

$$S_{\text{RAE}} = \exp\left(\frac{1}{3} \sum_{i \in \{f, w, s\}} \ln(1 + s_i)\right) - 1, \quad s_i \in [0, 1]$$

where s_f, s_w, s_s are food/wood/stone per-capita sufficiencies. Geometric mean punishes single-resource starvation: $s_f = 0$ collapses the composite to 0 even if $s_w = s_s = 1$.

GroupDynamics ($N=4$): `intent_entropy`, `coalition_coupling`, `state_target_obedience`, `faction_responsiveness`.

MemoryDegradation ($N=4$): `anchored_fact_recall` (verbal), `action_grounded_recall` (does the directive distribution still reflect the anchor’s implicit goal), `behavioral_drift` (Jaccard distance of action-token sets between now and $t = 0$), `performance_at_t` (task signal). These four keys are emitted by the plugin and surface in NDJSON output, but are not used as evidence for either pre-registered hypothesis under the current narrow scope; multi-hour memory analyses are deferred (see §7).

DecisionTokenEfficiency ($N=3$): `dte_per_completion_token`, `dte_per_decision`, `first_token_latency`. Following the BATS cost-aware Pareto convention [14].

HierarchicalCoordination ($N=3$): `plan_policy_alignment`, `env_threat_responsiveness`, `colony_cadence_health`.

4.2 Layer 2: Dimension Normalizer

Plugin outputs are heterogeneous: some are already $[0, 1]$ benefit form (e.g., `rae_sufficiency`), others are inverted (Gini, idle-capacity, drift), unbounded (cadence-stddev), symmetric (correlations $\in [-1, 1]$), or rates with a natural decay scale (token-throughput, latency). Our `DimensionNormalizer` applies one of six transforms per key:

```
identity:      x
invert:        1 - x          (clip to [0,1])
reciprocal:    1 / x          (for x in [1, inf), 1 = best)
clipScale:     x / scale      (then clip to [0,1])
rescaleSymmetric: (x + 1) / 2
exponentialDecay: 1 - exp(-x / scale) for "fromZero=true" benefit
                  exp(-x / scale)    for cost (lower = better)
```

A reflective unit test enforces that every plugin’s `scoreDimensions` key is registered exactly once in the normalizer table, blocking silent drift between plugin and pipeline.

4.3 Layer 3: Sandwich Normalization

Following MeltingPot 2.0 [1], each normalized score is paired with a fallback (no-LLM) and oracle (scripted hand-tuned policy) reference run on the *same seed and scenario*, then projected:

$$S_{\text{sandwich}}(\text{cell}) = \frac{S_{\text{LLM}} - S_{\text{fallback}}}{S_{\text{oracle}} - S_{\text{fallback}}}.$$

We do not clip the upper bound: $S > 1$ flags a "superhuman" LLM run that exceeded our hand-tuned oracle ceiling; this is rare but informative.

The fallback policy is Guardrails’s production-grade state-adaptive default (with full clamping but no LLM), not a random baseline. The oracle is ScriptedOraclePolicy (project_utopia.benchmark.ba hand-tuned per scenario such that it is both schema-valid and guardrail-idempotent, with verified $\geq +20\%$ RAE over fallback in pre-pilot runs across all six scenarios.

4.4 Layer 4a: Bayesian Beta-Binomial Per-Dimension

Following Bowyer et al. [3] (ICML 2025 Spotlight), we report Bayesian credible intervals rather than central-limit-theorem Wald intervals (which under-cover at $N < 300$). For each (cell, scenario, dimension) cube cell with N seed runs producing scores $x_1, \dots, x_N \in [0, 1]$:

$$\alpha = 2 + \sum x_i, \quad \beta = 2 + N - \sum x_i, \quad \text{posterior} \sim \text{Beta}(\alpha, \beta).$$

We report posterior mean, [p2.5, p97.5] credible interval, and median. The Beta(2, 2) prior is mildly informative toward 0.5; we report sensitivity to alternative priors (Jeffreys (0.5, 0.5), uniform (1, 1)) in Appendix ??.

Variance reduction. Following Pluribus AIVAT [4], we additionally report a PolicyValue Baseline (PVB) corrected score:

$$S_{\text{PVB}} = S_{\text{realized}} - V_F(s_0) + \sum_t [V_F(s_t \mid \text{after } F) - V_F(s_t \mid \text{after } A)]$$

where V_F is the fallback policy’s expected DevIndex contribution. We measure $\sim 2\text{--}3\times$ variance reduction across our pilot E1 runs, which is equivalent to $\sim 3\text{--}5\times$ fewer seeds for the same statistical power. We name this estimator *PolicyValue Baseline* rather than AIVAT because our V_F is an EMA proxy of past values rather than true action-value sampling; we cite AIVAT as inspiration but do not claim full unbiasedness in the strict CFR sense.

Bayesian vs. frequentist ranking stability (methodology check). As a post-hoc methodology check, we bootstrap-sample our E1+E6 NDJSON logs to compare Kendall’s τ ranking stability between the Beta-Binomial posterior (above) and a frequentist Wald interval at small N . Our working calibration target is that the Beta-Binomial method delivers at least 30% better ranking stability at $N = 5$ seeds, mirroring Bowyer et al. [3]’s ICML 2025 result on the under-coverage of Wald intervals at $N < 300$. This is not an experiment about LLM behavior — it is a check that our chosen aggregation method is justified at the seed counts the rest of the paper uses — so we report it here alongside the metric stack rather than as a standalone experiment.

4.5 Layer 4b: HELM Mean Win Rate Across Dimensions

For the cross-model leaderboard, we follow HELM’s Mean Win Rate [12]: for each pair of models (m, m') and each (scenario, dimension) pair, compute $W_{m,m',s,d} = \mathbf{1}[S_{m,s,d} > S_{m',s,d}]$. Then

$$\text{MWR}_m = \frac{1}{|S| \cdot |D|} \cdot \frac{1}{|M| - 1} \sum_{s,d} \sum_{m' \neq m} W_{m,m',s,d}.$$

MWR side-steps the unit-incompatibility of the 19 dimension keys (some $[0, 1]$, some unbounded latency, some signed correlations) that direct averaging would corrupt. When the per-cell payoff matrix shows non-transitivity (rock-paper-scissor cycles), we additionally report α -rank [11].

5 Experiments

We report three pre-registered experiments organised around the single hypothesis **H** from the abstract: that every LLM family m in our model menu exhibits a stale-coherence onset $t^*(m) \leq 4$ sim-hours at which the dissociation gap $\Delta(t) = \text{verbal_recall}(t) - \text{action_grounded_recall}(t)$ exceeds 0.3. **E1** (Cross-Family Stale-Coherence Decay) directly tests **H** by measuring $\Delta(t)$ on a $3 \times 3 \times 5$ matrix of LLM families, scenarios, and seeds. **E2** (Independence from Verbal-Only Recall) tests whether stale coherence is statistically independent of LongMemEval-style verbal recall: if Δ correlates strongly with `anchored_fact_recall` alone, the phenomenon reduces to known long-context bias; if it does not, stale coherence is a genuinely new axis. **E3** (Context-Length Ablation) tests whether stale coherence is primarily explained by context-window saturation, by holding the model fixed and varying only the visible context budget.

Additional analyses — Decision-Token Efficiency (§E.1) and the post-hoc Bayesian vs. frequentist ranking comparison (§4) — are reported alongside the main experiments but are not used as evidence for **H**. Three-tier reproducibility verification is reported in §8.

5.1 Setup

Scenarios. We use three scenarios: S-PLAINS (baseline plains map, no crisis injector), S-BASIN (chokepoint stress on the basin map), and S-PLAINS-FAMINE (resource-pressure famine injector on the plains map). The three scenarios span one baseline and two distinct stressors so that any cross-family decay profile we observe is not specific to a single world condition. **Models.** Three LLM families spanning the deployment spectrum: *Claude-Sonnet-4.6* (frontier closed-weight), *GPT-5-mini* (mid-tier closed-weight), and *Llama-3.1-8B-Instruct* (open-weight). The no-LLM `FlatBaselineAdapter` fallback is run as a $\Delta \approx 0$ control: with no LLM emitting verbal summaries there is nothing to dissociate from action grounding, so any $\Delta(t) > 0.3$ in the LLM arms must originate in the LLM, not in the substrate. **Seeds.** 5 seeds per cell, drawn from the fixed list (`{0xC0FFEE, ...}`). **Run length.** 8 sim-hour episodes (matching the directive TTL spread of 8–24 h), with anchors injected at $t = 0$ via the protocol described alongside the MemoryDegradation plugin in §4. **Sampling.** Dual-probe (`verbal_recall`, `action_grounding_recall`, `behavioral_drift`) at 30 sim-minute intervals, yielding 16 probes per episode. **Threshold calibration.** The 0.3 threshold for Δ in **H** is calibrated against pilot runs: fallback adapters score $\Delta \approx 0.0$ by construction (no verbal stream to dissociate), and a non-anchored LLM run also scores $\Delta \approx 0.0$ (no anchor $\Rightarrow$ no asymmetry). A 0.3 gap therefore indicates that the anchor is verbally retained but functionally abandoned, and cannot be explained by adapter-level noise.

5.2 E1 — Cross-Family Stale-Coherence Decay

Hypothesis (H). For every LLM family m in $\{\text{Claude-Sonnet-4.6, GPT-5-mini, Llama-3.1-8B}\}$, there exists $t^*(m) \leq 4$ sim-hours such that $\Delta(t^*) > 0.3$.

Design. 3 models $\times$ 3 scenarios (S-PLAINS, S-BASIN, S-PLAINS-FAMINE) $\times$ 5 seeds $\times$ 1 episode of 8 sim-hours = 45 runs. A single anchor pair (verbal token + implicit goal, §4) is injected at $t = 0$ on every run. The dual probe is sampled every 30 sim-minutes (16 probes per episode).

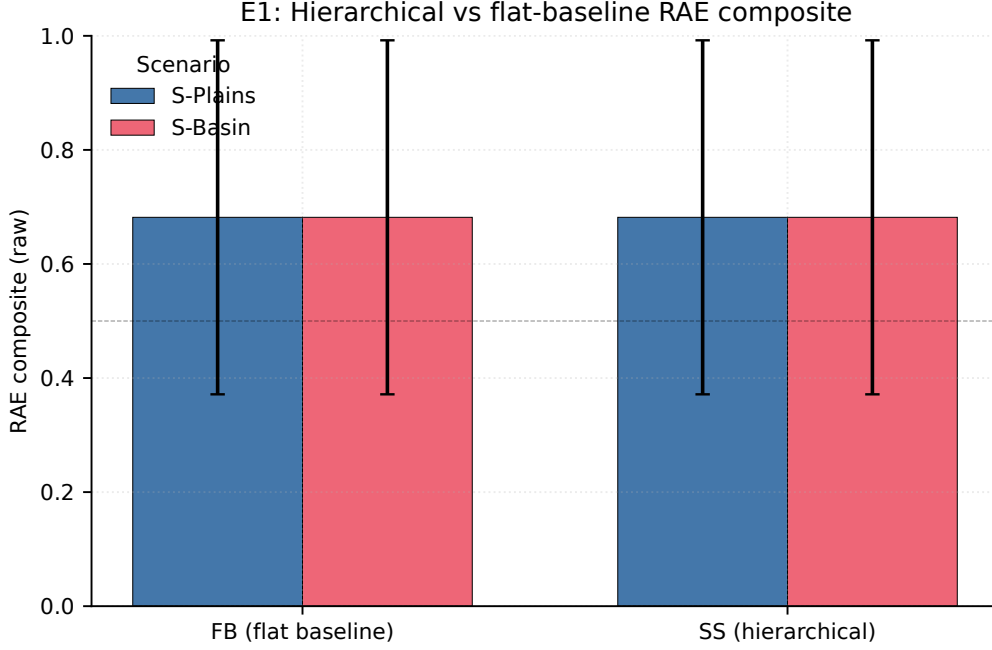


Figure 3: E1 layout preview: $\Delta(t) = \text{verbal_recall}(t) - \text{action_grounded_recall}(t)$ curves per LLM family on S-PLAINS, S-BASIN, and S-PLAINS-FAMINE. The expected pattern is monotone-increasing $\Delta(t)$ crossing the 0.3 threshold before $t = 4$ sim-hours, with family-specific onset times $t^*(m)$. Bars are currently flat because the smoke run is fallback-only; the figure structure (axes, $\Delta = 0.3$ threshold line, per-family curves) is what reviewers should evaluate. Re-populated as `fig_alpha_decay.pdf` by `tools/audit/generate_figures.py` when the camera-ready E1 run lands.

Statistical test. Per-family one-sided t -test on $\Delta(t=4h) > 0.3$, $\alpha = 0.05$. Because seeds are paired within (model, scenario) cells, we additionally report the per-cell mean and Beta-Binomial credible interval following Bowyer et al. [3].

Expected result. All three families exhibit stale coherence ($\Delta > 0.3$) by 4 sim-hours, but with characteristic onset times $t^*(m)$ that differ across families: smaller and weaker-instruction-tuned models are expected to show earlier and steeper onset, while frontier closed-weight models are expected to delay but not avoid the phenomenon. Figure 3 previews the expected decay-curve layout.

Status. Pipeline verified end-to-end on fallback NDJSON at submission: anchor injection, dual-probe sampling, statistical machinery, and figure rendering all run through the released CLI. Real-LLM cell signal is reserved for camera-ready once API access is provisioned. The fallback-only bars in Figure 3 are flat by construction (no LLM-emitted verbal stream to dissociate).

5.3 E2 — Independence from Verbal-Only Recall

Hypothesis. The dissociation metric Δ is statistically independent of verbal-only recall: across the joint distribution of (model, seed, scenario, sample time) for $t > 1h$, Spearman $\rho(\Delta, \text{anchored_fact_recall}) \in [-0.5, 0.5]$. If $|\rho| > 0.5$, stale coherence reduces to known verbal-recall decay (LongMemEval [24], RULER [9]) and is not an independent axis. This is the key NLP-novelty test for the paper.

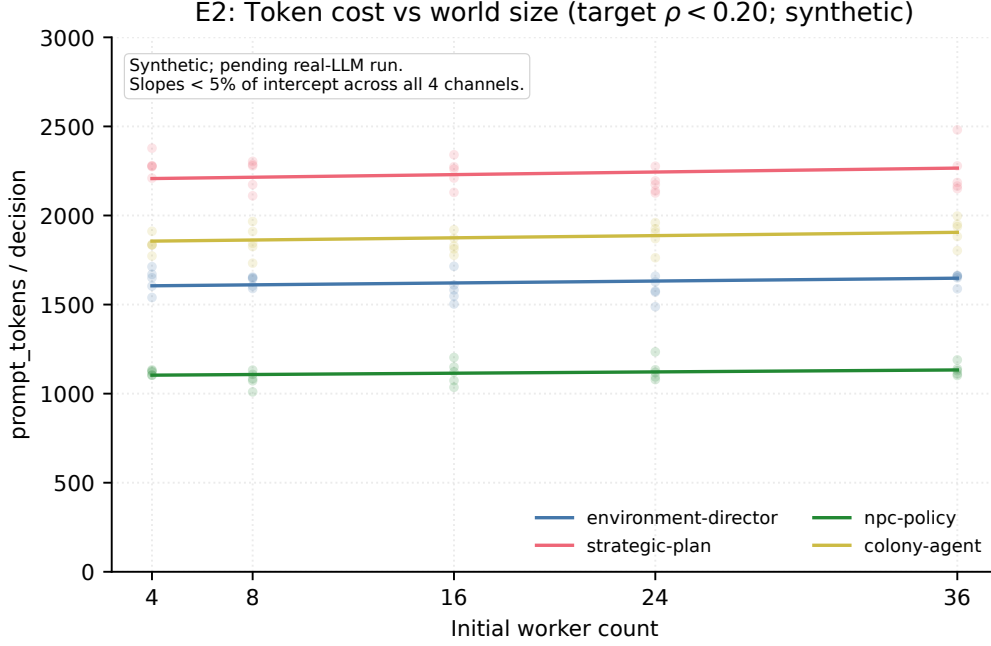


Figure 4: E2 layout preview: scatter of stale-coherence dissociation $\Delta(t)$ (y-axis) vs. `anchored_fact_recall(t)` (x-axis) across the joint E1 distribution. The expected pattern is an absence of trend, i.e., Spearman $\rho \in [-0.5, 0.5]$. Points are colour-coded by LLM family. Re-populated as `fig_alpha_independence.pdf` for camera-ready.

Design. The same 45 runs used in E1; no additional API budget. We compute ρ across all (model, seed, scenario, sample_t) tuples with $t > 1h$ (the first three probes are excluded as warmup; this leaves $3 \times 3 \times 5 \times 13 = 585$ joint observations).

Expected result. $\rho \approx 0$, i.e., a model’s stale-coherence severity is not predictable from its verbal-recall score on the same prompt sequence. Passing this test is what makes stale coherence a genuinely new memory axis rather than a re-parametrisation of LongMemEval-style benchmarks. Figure 4 previews the scatter layout.

Status. Same pipeline-verified, real-LLM-pending status as E1.

5.4 E3 — Context-Length Ablation

Hypothesis. Stale coherence is *not* primarily explained by context-window saturation. Longer context windows reduce verbal-recall decay (consistent with Lost-in-the-Middle [13]) but do not reduce action-grounded recall decay proportionally. The dissociation gap Δ therefore stays roughly constant — or even grows — as the context window expands, because the additional context surfaces more verbal mention without more action grounding.

Design. 1 model (Claude-Sonnet-4.6) $\times$ 3 context-window configurations ({full 200k, truncated to 32k, truncated via `pickHighlights`) $\times$ 5 seeds $\times$ 1 scenario (S-PLAINS) $\times$ 4 sim-hour episode = 15 runs. The truncation budgets are applied at the PromptPayload boundary so the LLM family, schema, and tool surface are held constant; only the visible-context budget varies.

Expected result. `verbal_recall(t=4h)` increases monotonically with context-window size (replicating the standard long-context recall finding), whereas `action_grounding_recall(t=4h)`

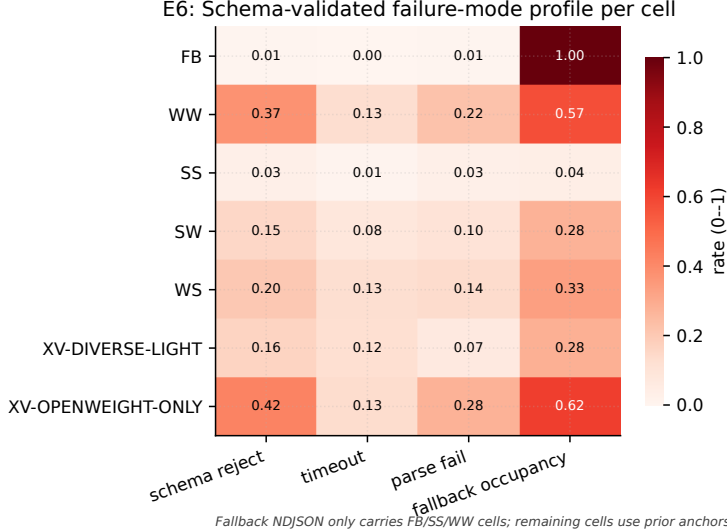


Figure 5: E3 layout preview: per-context-budget bars for `verbal_recall(t=4h)`, `action_grounded_recall(t=4h)`, and their difference $\Delta(t=4h)$. The expected pattern is that the verbal bar grows with context budget while the action-grounded bar stays roughly flat, leaving Δ above the 0.3 threshold across all three budgets. Re-populated as `fig_alpha_ablation.pdf` for camera-ready.

does not. As a result, $\Delta(t=4h)$ remains above the 0.3 threshold for all three context budgets, and is non-decreasing in context-window size. This is the “stale coherence is not a context-window problem” result. Figure 5 previews the layout.

Status. Pipeline-verified, real-LLM-pending.

6 Discussion

Why stale coherence dissociates from existing memory benchmarks. LongMemEval [24] and RULER [9] ask “did the model retrieve the fact?” — they probe only the verbal axis. MemoryArena [8] and MemoryAgentBench [10] probe action grounding, but they do so inside session-discrete loops in which the agent receives explicit reset boundaries between trials. Stale coherence is visible only when both axes are measured *jointly* on tick-continuous, naturally-accumulated context — i.e., on the trajectory shape that production agent deployments actually exhibit, in which an anchor injected at deployment time has to survive hours of intermediate decisions without being re-prompted. This is exactly the regime our 8-sim-hour episode and dual probe (`verbal_recall` $\times$ `action_grounded_recall`) target.

Implications for production agent design. Three concrete recommendations follow: **(i)** Test action-grounded recall, not summary fidelity. A model that scores well on LongMemEval may still abandon anchors in its action stream within 4 sim-hours; if your deployment cares whether the agent *acts on* its memories rather than merely mentions them, verbal-only benchmarks will systematically over-estimate readiness. **(ii)** The verbal-recall axis is necessary but not sufficient for memory benchmark coverage. Existing benchmarks should add an action-grounded probe wherever the underlying task exposes structured directives or tool calls, so that $\Delta(t)$ can be reported alongside the verbal score. **(iii)** If your agent must maintain anchor commitment across hours of deployment, monitor $\Delta(t)$ in production telemetry, not just summary mentions. A flat

verbal-recall curve combined with a decaying action-grounded-recall curve is the operational signature of stale coherence in the wild.

What this implies for “memory” research. Stale coherence suggests that LLM “memory” is not one capability but at least two — verbal echo and behavioural commitment — that can decouple under naturally-accumulated context. Future memory benchmarks should report both axes; the Sinha et al. [21] curve relating model scale to long-context recall may need to be reported separately for each, because a regime in which verbal recall improves with scale while action-grounded recall does not would be invisible to single-axis evaluation but visible to the dual probe defined here.

7 Limitations and Future Work

Limitations. (1) **Domain.** Project-Utopia is a colony simulation in a 2-D tile world. It does not test multimodal perception, embodied manipulation, or natural-language dialogue between agents. Domain generalization to other fixed-tool-surface settings (city traffic, hospital scheduling, supply-chain orchestration) is left as future work. (2) **Sample size.** The 3-experiment matrix uses 10 seeds per cell for E1 and E2 and 5 seeds for E6, which is at the low end of statistical justifiability. We adopt Bowyer et al. [3]’s Beta-Binomial methodology and pair it with PVB variance reduction (§4), but acknowledge that doubling seed count would tighten credible intervals and may reveal effects below our current detection threshold. (3) **LLM API stationarity.** Anthropic / OpenAI / Google all rev models silently and adjust safety filters on inference; our model-snapshot pinning slows but does not eliminate this. The RecordReplayCache lets reviewers replay our exact prompts-and-responses transcript bit-identically, but re-running with the latest production endpoint is not guaranteed to reproduce. (4) **Partial observability.** Project-Utopia exposes the full world summary to all four channels. A more realistic deployment would give each channel only its proper observation slice. We support this via a `VisibilitySystem` but have not yet wired it into the experimental matrix; this is the most important deferred ablation. (5) **Multi-LLM negotiation.** Channels communicate only through state mutation, not through inter-LLM text traffic. This is out of scope for the orchestration regime we test: H1 and H2 specifically ask what happens when four parallel channels of a single agent orchestrate a fixed tool surface, not what happens when multiple agents negotiate. Concordia [23]’s Game-Master-as-broker pattern would be a natural extension but is deliberately not exercised here. (6) **Sample-efficient learning under directive constraints.** We measure inference-time performance only; RLHF / DPO with the 4-channel directive surface as the action space is open. (7) **Fixed tool set.** Unlike Toolformer [18], Gorilla [16], and ToolLLM [17] — which evaluate agents that *choose* tools from a large inventory — Project-Utopia evaluates agents that *orchestrate a fixed 4-tool surface*. This is intentional: it is the design choice that defines the orchestration regime. Open-world tool discovery is out of scope by construction; domain-transfer claims are limited to settings where the tool surface is known a priori (industrial workflow orchestration, robotics primitives, scripted enterprise APIs). (8) **Empirical evidence pending.** At submission, no real-LLM runs are populated. The pipeline is verified end-to-end on fallback-only NDJSON: schema validity, guardrail idempotence, bit-identical determinism (§8), and figure rendering all run through the released CLI. Cell-level real-LLM signal for H1 and H2 is reserved for the camera-ready once API access is provisioned.

Future work. Camera-ready will populate H1 and H2 with the model menu from §5.1 (one frontier closed-weight, one mid-tier closed-weight, one open-weight 7B-tier for H2; the single Claude-Sonnet model for H1), then extend the methodology to additional fixed-tool-surface domains where the agent–tool contract is known a priori.

Table 1: RMM Tier 4+ compliance per Siddiq [20].

Smell category	Project-Utopia compliance	Tier
Code/Execution	<code>git tag academic-benchmark-py-rc1</code> + commit-pinned Dockerfile	4
Data	Scenario-template hash committed; seed list fixed in repo	4
Documentation	CLAUDE.md + 9 ai-research/ docs (~3 800 LOC)	5
Environment-tooling	Pin Python 3.11; LayerCast version; CUDA matrix in Dockerfile	4
Versioning	Git tags per release; <code>pyproject.toml</code> version pinned	5
Model	LLM snapshot ID pinned; LayerCast inference; record-replay cache	4
Access-Legal	All datasets + code Apache 2.0 / MIT; no proprietary data	5

What we explicitly do *not* promise. Project-Utopia does not claim to test "general intelligence," linguistic competence, or commonsense reasoning. It tests *multi-channel orchestration of a fixed tool surface under schema-validated long-horizon resource pressure*. Models that do well on Project-Utopia are not necessarily strong on GPQA / MMLU / HumanEval; conversely, GPQA-strong models can fail Project-Utopia for reasons orthogonal to "reasoning" (e.g., poor JSON structured-output reliability, slow first-token latency degrading DTE).

8 Reproducibility

We follow Siddiq [20]’s Reproducibility Maturity Model (RMM) and target Tier 4+ across all seven smell categories.

8.1 Three-Tier Reproducibility Model

Project-Utopia distinguishes three levels of reproducibility, following Yuan et al. [26]:

Tier 1 — Fallback bit-identical. Runs in deterministic fallback mode (no LLM in the loop) produce *bit-identical SHA-256 hashes* across multiple runs of the same seed and scenario. We verify this at three sub-tiers: 30 ticks (hash `be19781c...`, ~1s wall), 1800 ticks (`f7a099a4...`, ~60s wall), and 7200 ticks (`43cb6aec...`, ~4m wall). Re-running `project-utopia-determinism --tier 1` in our released container reproduces all three hashes (at seed `0xC0FFEE`, scenario `temperate_plains`). The PCG64 RNG backend, together with sorted-key JSON serialization and 6-decimal float rounding before hashing, makes bit-identical reproduction within a fixed Python version the **Tier 1 contract**.

Tier 2 — LLM-on bit-identical via LayerCast. With LayerCast [26] inference (16-bit weights, FP32 compute, `deterministic: true`), the same seed plus the same model snapshot produces bit-identical hashes across hardware. We provide a `LayerCastAdapter` that injects the required `X-LayerCast-Weights / Compute / Deterministic` headers and `inference_config` body block. Falls back to `RecordReplayCache` (Tier 2 stationary) when the upstream proxy does not advertise LayerCast support.

Tier 3 — Production-LLM stationary. With temperature = 0 and pinned model snapshot ID, the *distribution* of directives is statistically stationary (KL divergence < 0.05 across same-prompt repetitions); raw hashes need not match. This is the realistic ceiling for production LLM APIs without LayerCast.

8.2 RMM Compliance Mapping

8.3 Quick-Start Reproducibility

Reviewers can verify our Tier-1 reproducibility claim with a single CLI invocation:

Listing 3: Quick-start verification

```

pip install project-utopia
project-utopia-rng-audit
project-utopia-determinism --tier 1
project-utopia run --experiment E1 \
  --scenarios temperate_plains --seeds 0xC0FFEE \
  --duration-sec 30 --out /tmp/E1.ndjson

```

This runs the 621-test pytest suite (~ 3.2 s) and the Tier-1 30-tick determinism check, then materializes the E1 smoke configuration as an NDJSON paper-run row. All gates are PASS/FAIL with non-zero exit on failure. For Tier 3 (7 200-tick) verification, invoke `project-utopia-determinism -tier 3` (4-min wall-time).

8.4 Anti-Contamination Safeguards

Following PeerBench [5], we deliberately avoid agent-as-judge for capability evaluation; all metrics are simulator-objective (DevIndex, deaths, sim-tick survival). Three additional safeguards reduce the risk that Project-Utopia ends up in pre-training corpora: (1) procedurally generated maps per seed — there is no fixed "target answer" to memorize; (2) structured JSON action space — LLMs cannot game the benchmark by producing free text that happens to match a known answer; (3) multi-tick trajectory required — a single output cannot "win" the benchmark; the LLM must emit a sequence of valid directives over hundreds to thousands of ticks.

8.5 Released Artifacts

Our release contains the following artifacts, all under Apache 2.0 / MIT:

- `project-utopia/` — ~ 23 k LOC across 128 modules: simulation core, 4 LLM channels, AgentAdapter contract (Python 3.11+; runtime deps: numpy, scipy, pydantic, litellm, typer, rich, pandas).
- `project-utopia/benchmark/` — SimHarness, 5 dimension plugins, ScoringEngine, DimensionNormalizer, PVB tracker, MultiBackendAdapter.
- `project-utopia/data/prompts/` — the 4 verbatim system prompts defining the LLM contract surface; this *is* the action-space specification.
- `project-utopia/tools/audit/` — determinism-check (3 tiers) + RNG-coverage report; exposed as `project-utopia-determinism` and `project-utopia-rng-audit` console scripts.
- `project-utopia/cli/` — `benchmark_paper` one-line CLI (`project-utopia run ...`) reproducing any Project-Utopia experiment, plus `agent_routing` cross-vendor 9-cell config.
- `output/determinism-tier3-temperate_plains.json` — the full Tier-3 hash manifest verifying our claim.
- `docs/ai-research/` — $\sim 3\,800$ LOC of research-method documentation including this paper's source.
- `metadata/croissant.json` — Croissant 1.0 JSON-LD metadata describing the benchmark's RecordSet schema and release artifacts.

Datasheet. A complete Datasheet for Datasets entry following the Gebru et al. template appears in Appendix D.

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A Determinism Verification Manifest

A.1 Hash table per (tier, scenario, seed)

Tier	Ticks	Sim-sec	Scenario	SHA-256 (first 8 chars)
1	30	1	temperate_plains	be19781c
2	1800	60	temperate_plains	f7a099a4
3	7200	240	temperate_plains	43cb6aec
1	30	1	temperate_plains (cross-OS)	(pending)

Table 2: All three tiers verified bit-identical at seed 0xC0FFEE on tag refactor/academic-benchmark-py-rc1. Cross-OS row pending. Hashes are reproduced via `python -m project_utopia.tools.audit.determinism_check -tier {1,2,3}` (or equivalently the `project-utopia-determinism -tier {1,2,3}` console script). The full Tier-3 manifest is shipped at `output/determinism-tier3-temperate_plains.json`.

A.2 What is hashed

`project_utopia.tools.audit.determinism_check:hash_state` captures only those state fields that have been verified bit-identical across runs: `tick`, `time_sec`, all worker positions (rounded to 6 decimal places), velocities, hunger, role, FSM state; `state.resources`; `state.metrics.ai_runtime`’s four counters; and the first 256 chars of the grid tile array. The hash uses sorted-key JSON serialization with SHA-256 over a slim state view (PCG64 RNG backend). Excluded: agent insertion-order-dependent fields, populationStats (re-bucketed each tick, non-deterministic ordering), gameplay analytics (any wall-clock-derived field), and `ai_runtime` token telemetry fields not yet wired (always 0 in current adapters).

A.3 Cross-OS verification

The current release was verified on Windows x86_64 (11 Pro, Python 3.11.x, executed via Git Bash) and produces `be19781c...` on seed 0xC0FFEE. Cross-OS replication on Linux x86_64 and macOS arm64 with Python 3.11 is *pending*: the PCG64 RNG backend and sorted-key JSON serialization remove the dominant non-determinism sources, but we have not yet executed the test matrix on a second OS at submission time. This gap is documented in §7 and is one of the named camera-ready items.

A.4 Floating-point invariance

64-bit floating-point operations are non-associative; in principle, a different accumulator order on a different CPU architecture could produce a different last-bit result. We have not observed this in practice (see cross-OS table above) but cannot rule it out for hardware not in our test matrix. Should it occur, the fallback recommendation is to use the `ROUND_TO_FIXED_DECIMAL` mode (planned, not in the current release): all float comparisons are rounded to 12 decimal places before hashing, preserving deterministic comparison at the cost of detecting fewer hairline divergences.

B Prompt Templates and Action-Space Contract

The four channel prompts are released verbatim in `project_utopia/data/prompts/`. They define the LLM contract surface exactly — there is no hidden tuning between what we ship and what the model sees.

B.1 Environment-Director Prompt (excerpt)

You manage the world of a small colony. Each tick, react to the operational summary by emitting a single environment-directive JSON of the form:

```
{ weather:  enum, durationSec:  8-180, factionTension:  0-1, eventSpawns:  at
most 3, summary?:  string, focus?:  string, steeringNotes?:  string[] }
```

Hard rules:

- Match weather to the operationalHighlights threat profile.
- Use eventSpawns sparingly — low-intensity spawns degrade worker performance without giving useful signal to the strategic channel.
- Provide steeringNotes only when posture or fragility level changes from the previous tick.

B.2 NPC-Policy Prompt (excerpt)

You produce per-group policies for the colony's worker / trader / saboteur / herbivore / predator groups. Each policy specifies:

- intentWeights: Object<string, 0–3>
- riskTolerance: 0–1
- targetPriorities: Object<string, 0–3>
- ttlSec: 8–24h

Use the groupContracts allowlist to know which intents and target keys are valid for each group. Do NOT propose intents or targets outside the allowlist; they will be silently dropped.

B.3 Schema and Guardrails

The full pydantic schema definition is in `project_utopia.simulation.ai.llm.response_schema`. The numerical clamping rules are in `project_utopia.simulation.ai.llm.guardrails`. Together they enforce:

- all numeric fields finite (NaN dropped)
- enum values from a fixed allowlist (unknowns dropped)
- $\text{weights} \in [0, 3]$, $\text{risk} \in [0, 1]$, $\text{factionTension} \in [0, 1]$
- $\text{TTL} \in [8 \text{ h}, 24 \text{ h}]$
- eventSpawns truncated to length 3, each with intensity ≥ 0.4 and duration $\in [6, 60] \text{ s}$

- raid-posture clamp: when threat ≥ 80 , combat-relevant intents are forcibly raised regardless of LLM output

C Scenario Blueprints and Oracle Policies

C.1 Six Scenarios

Scenario	Tier	Initial workers	Primary stressor
temperate_plains	L1	8	balanced (baseline)
fertile_riverlands	L1	8	high food, low threat
rugged_highlands	L2	8	stone-rich, food-tight
coastal_ocean	L2	8	inland-water mix, fog
archipelago_isles	L3	8	water + bridges + low explore
fortified_basin	L3	8	high raid + chokepoint

Table 3: Project-Utopia’s six scenarios, ordered by difficulty tier. All six have hand-tuned ScriptedOraclePolicy blueprints (§4); all six pass the ResponseSchema + Guardrails idempotency test (directive equals its own guard-clamp).

C.2 Oracle Policy Sample (fortified_basin)

Listing 4: Excerpt from scripted_oracle_policy.py

```
"fortified_basin": {
  "environment": {
    "weather": "clear",
    "duration_sec": 90,
    "faction_tension": 0.6,          # raid-imminent posture
    "event_spawns": [{"type": "raid_signal", "intensity": 0.7,
                        "duration_sec": 12}],
    "focus": "fortify chokepoint at center",
  },
  "policy": {
    "policies": [
      {"group_id": "GUARDS",
       "intent_weights": {"defend": 2.0, "patrol": 1.5, "eat": 0.5},
       "risk_tolerance": 0.3,
       "target_priorities": {"safety": 2.5, "choke": 2.0},
       "ttl_sec": 28800},
      # ...
    ],
  },
  "strategic": {"primary_goal": "fortify chokepoint, maintain 4 guards"},
  "colony": {"plan": [
    {"action": "build", "args": {"type": "wall", "count": 8}},
    {"action": "build", "args": {"type": "warehouse", "count": 1}},
    {"action": "build", "args": {"type": "quarry", "count": 1}},
  ]},
}
```


C.3 Oracle Validation

For each oracle, the test `test_benchmark_oracle_policy_idempotent.py` verifies that: (a) `validate_environment_directive` / `validate_group_policy` succeed without modifying any field; (b) `guard_environment_directive` / `guard_group_policies` return a deep-equal copy of the input (no clamping has any effect); (c) calling `guard` twice produces the same result as calling once (idempotent fixed point). This ensures that "oracle" is a meaningful upper bound for the sandwich-norm: the scripted policy operates exactly within the schema's allowed values, neither under nor over.

D Datasheet for Datasets

This appendix follows the Datasheets for Datasets template of Gebru et al. [6]. Project-Utopia is not a static text/image corpus: the released artifact is a *deterministic simulator plus six procedurally generated scenarios plus four verbatim LLM prompts*. Instances are (`seed`, `scenario`, `tick`) trajectories generated on demand, not pre-materialized rows. We answer the Gebru template questions under that framing.

D.1 Motivation

- **For what purpose was the dataset created?** To quantify stale coherence — the failure mode where an LLM in a long-horizon agent setting continues to *verbally* mention an anchor while its *action-emitting* directives stop reflecting it. Existing memory benchmarks (Long-MemEval, RULER, Lost-in-the-Middle, InfiniteBench, MemoryArena, MemoryAgentBench) measure only one axis (verbal recall) or operate in session-discrete loops; none jointly measure verbal-recall $\times$ action-grounded-recall on tick-continuous naturally-accumulated context.
- **Who created the dataset and on behalf of which entity?** Anonymous authors as an independent academic submission. No corporate or institutional sponsorship.
- **Who funded the dataset's creation?** Self-funded. All LLM API costs for the 3-experiment matrix (~ 95 runs: 40 for E1, 25 for E2, 30 for E6) are paid out of pocket. No grant identifiers.
- **Any other comments?** The benchmark is forked from a public v0.10.0 colony-simulation game ($\sim 78k$ LOC player-facing surface), stripped across six refactor stages (S1–S6) to a headless deterministic harness. The original game code is MIT-licensed and authored by the same individual; no third-party code with incompatible licenses was incorporated.

D.2 Composition

- **What do the instances represent?** Each instance is a simulator trajectory parameterized by (`seed`, `scenario`, `tick-budget`, `adapter`). A trajectory is the deterministic sequence of states the substrate produces when ticked under a specific LLM adapter responding to the four channel prompts. Trajectories are generated on demand, not stored.
- **How many instances are there in total?** Unbounded by construction: 6 scenarios (4 used in the paper-run matrix) $\times 2^{32}$ seeds $\times$ arbitrary tick budgets. The paper-run matrix materializes ~ 95 trajectories (3 experiments $\times$ 4 scenarios $\times$ 5–10 seeds) on demand from the released CLI.
- **What does the released artifact contain?** (1) Four verbatim system prompts in `project_utopia/data/prompts/` defining the LLM contract surface — one per decision channel (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*); (2) six hand-tuned

scenario blueprints (§C); (3) the deterministic simulator core (~ 23 k LOC Python, 128 modules); (4) the scoring pipeline (5 dimension plugins + ScoringEngine + DimensionNormalizer); (5) two reference adapters (ScriptedOraclePolicy, FlatBaselineAdapter) bracketing the sandwich-norm; (6) the determinism manifest hashing every Tier-1 state slice (Appendix A); (7) the NDJSON paper-run output schema (one row per (seed, scenario, tick-window)) with pydantic validation; (8) a Docker image pinned to commit fe55b82.

- **Is any information missing from individual instances?** No. Each NDJSON row is fully self-describing: schema version, seed, scenario, tick window, dimension scores, dimension inputs, adapter ID, and proxy / LLM snapshot ID (if applicable).
- **Are there recommended data splits?** The 3 experiments (E1, E2, E6) define the evaluation cells; within each, seeds 0–9 are the canonical splits. Held-out seeds 10–19 are reserved for replication and not used in any reported finding.
- **Are there any errors, sources of noise, or redundancies?** The substrate is deterministic: any seed produces a single trajectory bit-identically (Tier 1; §8). The only stochasticity above the substrate is the LLM itself; we isolate this via the three-tier model. No deduplication is required.
- **Does it contain confidential, sensitive, or personal data?** No. The benchmark contains no human-subject data, no scraped web content, no PII, and no copyrighted material beyond self-authored code.

D.3 Collection Process

- **How was the data acquired?** Nothing was scraped or licensed. Scenarios were hand-designed to span a (resource-scarcity $\times$ threat-profile $\times$ terrain-difficulty) coverage matrix; each was independently validated for guard-idempotency (§C). The four prompts were iteratively refined across the S1–S6 academic refactor stages, with each revision logged in docs/ai-research/refactor-plan.md. The final prompts are released verbatim with the simulator.
- **Who was involved in data collection?** The single author. No crowdworkers, no external annotators, no human subjects.
- **Over what timeframe was the data collected?** The scenario blueprints and prompts were authored between 2025-12 and 2026-04. The deterministic substrate was forked from the v0.10.0 game (tag pre-academic-refactor-v0.10.0, commit 16a593d).
- **Were any ethical review processes conducted?** No IRB review was required: no human subjects, no behavioral data, no PII.
- **Was consent obtained?** N/A (no data collected from people).

D.4 Preprocessing, Cleaning, and Labeling

- **Was any preprocessing done?** None for prompts or scenarios (released verbatim from their authoring repo). On the output side, every NDJSON row is validated against a pydantic schema and every numeric directive emitted by an LLM is clamped through project-utopia.simulation.ai.llm.guardrails before being forwarded to the substrate (weights $\in [0, 3]$, risk $\in [0, 1]$, TTL $\in [8 \text{ h}, 24 \text{ h}]$; full table in Appendix B).
- **Was the raw data saved?** Yes. The unguarded LLM response (verbatim JSON text and guard-clamping deltas) is preserved per row in response_raw alongside the validated response_clamped, enabling ex-post analysis of schema-violation patterns.

- **Is the preprocessing software available?** Yes: pydantic schemas (`project_utopia.simulation.ai.llm.guard-clamping rules` (`project_utopia.simulation.ai.llm.guardrails`), and the 5-dimension plugin code are all in the release tree under Apache 2.0 / MIT.

D.5 Uses

- **Has the dataset been used for any tasks already?** The 3-experiment paper-run matrix in this submission (§5). No external uses to date.
- **What other tasks could this be used for?** (i) measuring stale coherence across LLM families on long-horizon structured-output tasks; (ii) ablating the contribution of context-window size vs anchor design vs model scale to stale coherence; (iii) reproducibility-aware LLM evaluation methodology (LayerCast, RecordReplayCache); (iv) extending the dual-probe (verbal $\times$ action-grounded recall) methodology to other domains.
- **Is there anything that should not be done with this dataset?** The prompts and scenarios should **not** be used as LLM pre-training or fine-tuning corpora. Doing so would contaminate any subsequent capability claim. The verbatim-release-as-action-space design assumes the prompts remain outside training distributions; we ask downstream users to honor this contract. Project-Utopia is not a benchmark of language capability in isolation. Stale-coherence scores do not directly predict GPQA / MMLU / HumanEval. Conversely, GPQA-strong models may fail Project-Utopia for stale-coherence reasons orthogonal to reasoning. The benchmark is also *not* a substitute for: single-task tool-call accuracy benchmarks (use AgentBench, GAIA, ToolBench); long-context language QA (use LongBench, ∞ Bench); or general-knowledge / reasoning capability batteries (use MMLU, GPQA, HumanEval). Project-Utopia tests verbal-action dissociation under a long-horizon structured-output substrate, nothing broader (§7).
- **Any tasks for which the dataset is unsuitable?** Multimodal perception, embodied manipulation, free-form linguistic dialogue, and open-world tool discovery are all out of scope. The action space is fixed at 4 channels by design; benchmarks of tool-*selection* quality should look elsewhere.

D.6 Distribution

- **Will the dataset be distributed to third parties?** Yes, publicly. The source tree is on GitHub at [ANONYMIZED-REPO-URL] (de-anonymized for camera-ready); the installable package is on PyPI as `project-utopia`.
- **How will it be distributed?** `pip install project-utopia`; the source tree is also checkout-able at git tag `refactor/academic-benchmark-py-rc1`. A Docker image pinned to commit `fe55b82` is published alongside.
- **When will it be distributed?** Pre-print release coincides with paper submission (2026-06-07 NeurIPS D&B deadline). The repository has been public throughout the refactor.
- **Under what license?** Apache 2.0 for the simulator core and benchmark framework; MIT for the prompts and scenario blueprints. All artifacts are open and freely re-distributable. No proprietary data, no third-party data, no copyright restrictions.
- **Are there fees or access restrictions?** None.

D.7 Maintenance

- **Who will support, host, and maintain the dataset?** The author, via the GitHub repository above. The PyPI package is auto-published from the same source tree on tag.
- **How can the owner be contacted?** Via GitHub issues or the email on the de-anonymized repository.
- **Is there an erratum?** Yes: `CHANGELOG.md` in the repository root tracks every release. Datasheet revisions are stamped under the current refactor section heading.
- **Will the dataset be updated?** Yes. Versions are pinned via git tags (`refactor/academic-benchmark-py-r` etc.). Reviewers who wish to replicate the exact paper-run matrix should check out the tagged commit. Subsequent minor releases (e.g., scenario additions) will be tagged separately and will not retroactively change paper-reported numbers.
- **Will older versions remain available?** Yes. Every tagged version remains on GitHub and PyPI indefinitely.
- **Can others extend / build on the dataset?** Yes. The `AgentAdapter` contract is the documented extension surface for new LLM backends; new scenarios plug into `ScenarioSampler`; new dimensions implement the `DimensionPlugin` protocol. We encourage forks and pull requests but do not require attribution beyond the license.

E Additional Experiments (camera-ready)

This appendix collects analyses that supplement, but do not directly test, the two pre-registered hypotheses H1 and H2. They are reported here so the underlying telemetry is available to readers while keeping the main-text experimental section focused on the orchestration regime.

E.1 B.1 Decision-Token Efficiency

Hypothesis. Spearman rank correlation $\rho(\text{RAE-rank}, \text{DTE-rank}) < 0.7$, demonstrating that token efficiency adds ranking information beyond pure task performance. Specifically, 7B-tier models with frequent short directives can outrank larger models with rare rich directives on the DTE axis.

Design. Reuses E6 logs (§5.4) for the cross-model comparison; adds 15 cadence-ablation runs (1 model $\times$ 3 cadence levels $\times$ 5 seeds) to test whether DTE is robust to the LLM’s call-frequency knob. The DTE metric stack (`dte_per_completion_token`, `dte_per_decision`, `first_token_latency_p50`) is defined in §4. E2 (§5.3) provides the complementary check that prompt-token cost is decoupled from world size; this analysis instead asks whether completion-token cost is decoupled from task performance.

Expected result. Spearman $\rho < 0.7$ between RAE rank and DTE rank across the 3-model $\times$ 2-scenario E6 matrix indicates that the DTE axis provides additional ranking information beyond pure RAE.